## Investigation: METALGRAFICA IGUACU S.A. (CVM code 8605), fiscal year 2011, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
Trading codes (2022 FCA, version 5, filed on 2022-10-25): MITG03 (Ações Ordinárias, Bolsa, since 1968-09-02); MITG04 (Ações Preferenciais, Bolsa, since 1968-09-02).

## Document used

2011 DFP, version 1, filed on 2012-03-23; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2012-03-23).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=15279

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: component not computable: DEPI
- Alert by the study 1 rule: does not trigger alerts
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

DSRI, days' sales in receivables index
- Value: 1.998 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 98 (n = 23)
- Percentile in group and year: 93 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.655 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 89 (n = 23)
- Percentile in group and year: 95 (n = 477)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.100 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 72 (n = 23)
- Percentile in group and year: 74 (n = 472)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 0.747 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 2 (n = 23)

Percentile in group and year: 8 (n = 478)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: missing account: depreciation, cash flow lines in section 6.01.01
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

SGAI, SG&A expenses index
Value: 1.403 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 93 (n = 23)
Percentile in group and year: 90 (n = 471)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.027 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 46 (n = 23)
Percentile in group and year: 57 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 0.845 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 11 (n = 23)
Percentile in group and year: 13 (n = 486)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: −25.3 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 35.1 days)
Percentile in sector and year: 2 (n = 22)
Percentile in group and year: 7 (n = 449)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 23.50 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 80 (n = 23)
Percentile in group and year: 82 (n = 438)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.026 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 54 (n = 23)
Percentile in group and year: 59 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 4.204 (index)
Components: X1 = 0.373; X2 = 0.012; X3 = −0.132; X4 = 2.480 (indices)
Percentile in sector and year: 73 (n = 24)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 5/9 (8 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 1
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 7 accounts above the R$1,000 tolerance (group a, totals and key lines: 1; group b, reclassifications: 6), in the annex at the end.

## What was known on the date

- DFP 2010, v1: 2011-03-25
- DFP 2010, v2: 2011-03-25
- FCA 2011, v1: 2011-03-25

- FCA 2011, v2: 2011-05-05
- DFP 2011, v1: 2012-03-23
- DFP 2012, v1: 2013-03-21

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Com Ressalva (qualified).
Excerpt checked literally: "Em nossa opinião, exceto quanto aos efeitos correspondentes ao assunto descrito no parágrafo sobre a base para opinião com ressalva, as demonstrações contábeis acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da Metalgráfica Iguaçu S.A. em 31 de dezembro de 2011 e 2010, o desempenho de suas operações, os seus fluxos de caixa e o valor adicionado para os exercícios findos naquelas datas, de acordo com as práticas contábeis adotadas no Brasil.Outros AssuntosConforme mencionado na Nota 16, a Companhia possui créditos a recebera Kremon do Brasil S.A."
Auditor in the 2020 FCA: Martinelli Auditores.
Auditor in the 2021 FCA: Martinelli Auditores.
Auditor in the 2022 FCA: Consult Auditores Independentes, Martinelli Auditores.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2014: Morre Stephens Boeing Auditores Independentes S/S -> Mazars Auditores Independentes - Sociedade Simples Ltda. The outgoing firm audited 5 consecutive fiscal years (2009 to 2013), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2017: entry of Morre Stephens Boeing Auditores Independentes S/S, with no other firm leaving.
- 2018: Mazars Auditores Independentes - Sociedade Simples Ltda -> Martinelli Auditores. The outgoing firm audited 3 consecutive fiscal years (2014 to 2016), by the start and end dates in the FCA.
- 2018: exit of Morre Stephens Boeing Auditores Independentes S/S, with no other firm entering. The FCA does not have the dates needed to count the outgoing firm's fiscal years.
- 2022: entry of Consult Auditores Independentes, with no other firm leaving.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why

the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:38:15+00:00.

## Today's information (outside the reference date)

Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

## Annex: full list of differences in the following year's comparative (R$ million)

(a) DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: 2.9 in the version used, 3.0 in the comparative (difference of 0.1)
(b) DFC (indirect method) 6.01.02.04 Impostos e Contribuições a Recuperar: 4.4 in the version used, 4.6 in the comparative (difference of 0.1)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: −1.0 in the version used, −0.9 in the comparative (difference of 0.1)
(b) DFC (indirect method) 6.03 Caixa Líquido Atividades de Financiamento: 1.7 in the version used, 1.6 in the comparative (difference of −0.1)
(b) DFC (indirect method) 6.01.02.08 Fornecedores: −6.9 in the version used, −6.8 in the comparative (difference of 0.1)
(b) DFC (indirect method) 6.01.02.12 Outras Exigibilidades: −0.8 in the version used, −0.9 in the comparative (difference of −0.1)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: 3.9 in the version used, 3.8 in the comparative (difference of −0.0)